

Statistical Reasoning HW1

Alex Melnichenko

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(a)

$$\begin{aligned}
 p(\sigma^{-2}|\mu = 0, y_1, \dots, y_n) &\propto p(y_1, \dots, y_n|\mu = 0, \sigma^2)p(\mu = 0)p(\sigma^{-2}) \\
 &= p(y_1, \dots, y_n|\mu = 0, \sigma^2)p(\sigma^{-2}) \\
 &\propto \prod_{i=1}^n \frac{1}{\sigma} \exp\left(-\frac{1}{2}\sigma^{-2}y_i^2\right) \cdot (\sigma^{-2})^{a-1} \exp(-b\sigma^{-2}) \\
 &\propto \sigma^{n+a-1} \exp\left(-\sigma^{-2}\left(\frac{1}{2}\sum_{i=1}^n y_i^2 + b\right)\right) \\
 \implies \sigma^{-2}|\mu = 0, y_1, \dots, y_n &\sim GAM\left(n+a, \frac{1}{2}\sum_{i=1}^n y_i^2 + b\right)
 \end{aligned}$$

I initially attempted to compute the distribution without given b though I couldn't tell if it was proportional to any known distribution, so left it as a proportion below.

$$\begin{aligned}
 p(\sigma^{-2}|\mu = 0, y_1, \dots, y_n) &\propto p(y_1, \dots, y_n|\mu = 0, \sigma^2)p(\mu = 0)p(\sigma^{-2}) \\
 &= p(y_1, \dots, y_n|\mu = 0, \sigma^2)p(\sigma^{-2}) \\
 &= p(y_1, \dots, y_n|\mu = 0, \sigma^2) \int_0^\infty p(\sigma^{-2}|b)p(b)db \\
 &= p(y_1, \dots, y_n|\mu = 0, \sigma^2) \int_0^\infty \frac{b^a}{\Gamma(a)}(\sigma^{-2})^{a-1}e^{-\sigma^{-2}b} \frac{\beta^\alpha}{\Gamma(\alpha)}b^{\alpha-1}e^{-b\beta}db \\
 &= p(y_1, \dots, y_n|\mu = 0, \sigma^2) \int_0^\infty \frac{b^a}{\Gamma(a)}(\sigma^{-2})^{a-1}e^{-\sigma^{-2}b} \frac{\beta^\alpha}{\Gamma(\alpha)}b^{\alpha-1}e^{-b\beta}db \\
 &= p(y_1, \dots, y_n|\mu = 0, \sigma^2) \frac{(\sigma^{-2})^{a-1}}{\Gamma(a)\Gamma(\alpha)} \frac{\Gamma(a+\alpha)}{(\beta + \sigma^{-2})^{a+\alpha}} \int_0^\infty \frac{(\beta + \sigma^{-2})^{a+\alpha}}{\Gamma(a+\alpha)} b^{a+\alpha-1} \exp(-b(\beta + \sigma^{-2}))db \\
 &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\sigma^{-2}y_i^2\right) (\sigma^{-2})^{a-1} \frac{1}{\Gamma(a)\Gamma(\alpha)} \frac{\Gamma(a+\alpha)}{(\beta + \sigma^{-2})^{a+\alpha}} \\
 &= \frac{1}{(\sqrt{2\pi})^n} \exp\left(-\frac{1}{2}\sigma^{-2}\sum_{i=1}^n (y_i)^2\right) (\sigma^{-2})^{\frac{n}{2}+a-1} \frac{1}{\Gamma(a)\Gamma(\alpha)} \frac{\Gamma(a+\alpha)}{(\beta + \sigma^{-2})^{a+\alpha}} \\
 &\propto \exp\left(-\frac{1}{2}\sigma^{-2}\sum_{i=1}^n (y_i)^2\right) (\sigma^{-2})^{\frac{n}{2}+a-1} \frac{1}{(\beta + \sigma^{-2})^{a+\alpha}}
 \end{aligned}$$

(b)

$$\begin{aligned}
p(b|a, \sigma^{-2}) &\propto p(\sigma^{-2}|a, b) \cdot p(a, b) \\
&= p(\sigma^{-2}|a, b) \cdot p(b) \\
&= b^{\alpha-1} \exp(-b\beta) (\sigma^{-2})^{a-1} \exp(-\sigma^{-2}b) \frac{b^a}{\Gamma(a)} \frac{\beta^\alpha}{\Gamma(\alpha)} \\
&\propto b^{\alpha-1} \exp(-b\beta) \exp(-\sigma^{-2}b) b^a \\
&= b^{\alpha+a-1} \exp(-b(\beta + \sigma^{-2})) \\
\implies b|a, \sigma^{-2} &\sim \text{GAM}(\alpha + a, \beta + \sigma^{-2})
\end{aligned}$$

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(a)

$$\begin{aligned}
P(\theta_1, \dots, \theta_K | y_1, \dots, y_n) &= P(\theta_1, \dots, \theta_K) \cdot P(y_1, \dots, y_n | \theta_1, \dots, \theta_K) \\
&\propto \prod_{k=1}^K (\theta_k^{\alpha_k-1}) \prod_{j=1}^n \prod_{k=1}^K (\theta_k^{I(y_j=k)}) \\
&= \prod_{k=1}^K (\theta_k^{\alpha_k-1}) \prod_{k=1}^K (\theta_k^{\sum_{j=1}^n I(y_j=k)}) \\
&= \prod_{k=1}^K (\theta_k^{\alpha_k-1} \theta_k^{\sum_{j=1}^n I(y_j=k)}) \\
&= \prod_{k=1}^K (\theta_k^{n_k + \alpha_k - 1}) \\
\implies \theta_1, \dots, \theta_K | Y_1, \dots, Y_n &\sim \text{DIR}(\alpha_1 + n_1, \dots, \alpha_K + n_K)
\end{aligned}$$

(b)

We note that $p(\theta_1, \dots, \theta_K) = \frac{\Gamma(\sum_{k=1}^K \alpha_k + n_k)}{\prod_{k=1}^K \Gamma(\alpha_k + n_k)} \prod_{k=1}^K (\theta_k^{n_k + \alpha_k - 1})$ is the density function of a Dirichlet distribution so we have $\int \dots \int_{\Theta} \frac{\Gamma(\sum_{k=1}^K \alpha_k + n_k)}{\prod_{k=1}^K \Gamma(\alpha_k + n_k)} \prod_{k=1}^K (\theta_k^{n_k + \alpha_k - 1}) d\theta_1 \dots d\theta_K = 1$, which we use getting to the final line.

$$\begin{aligned}
p(y_1, \dots, y_n) &= \int \dots \int_{\Theta} p(y_1, \dots, y_n | (\theta_1, \dots, \theta_K)) p(\theta_1, \dots, \theta_K) d\theta_1 \dots d\theta_K \\
&= \int \dots \int_{\Theta} \frac{\Gamma(\sum_{k=1}^K \alpha_k)}{\prod_{k=1}^K \Gamma(\alpha_k)} \prod_{k=1}^K (\theta_k^{n_k + \alpha_k - 1}) d\theta_1 \dots d\theta_K \\
&= \frac{\Gamma(\sum_{k=1}^K \alpha_k)}{\prod_{k=1}^K \Gamma(\alpha_k)} \cdot \left(\frac{\Gamma(\sum_{k=1}^K \alpha_k + n_k)}{\prod_{k=1}^K \Gamma(\alpha_k + n_k)} \right)^{-1} \cdot \int \dots \int_{\Theta} \frac{\Gamma(\sum_{k=1}^K \alpha_k + n_k)}{\prod_{k=1}^K \Gamma(\alpha_k + n_k)} \prod_{k=1}^K (\theta_k^{n_k + \alpha_k - 1}) d\theta_1 \dots d\theta_K \\
&= \frac{\Gamma(\sum_{k=1}^K \alpha_k)}{\prod_{k=1}^K \Gamma(\alpha_k)} \cdot \left(\frac{\Gamma(\sum_{k=1}^K \alpha_k + n_k)}{\prod_{k=1}^K \Gamma(\alpha_k + n_k)} \right)^{-1}
\end{aligned}$$

(c)

We have that $Y_{n+1} \sim CAT(\lambda_1, \dots, \lambda_K)$ where $(\Lambda_1, \dots, \Lambda_K) \sim DIR(\alpha_1 + n_1, \dots, \alpha_K + n_K)$ by (a). I'm not able to follow the notation like at all if I use θ here again hence a different variable. We additionally apply (b) to compute.

$$\begin{aligned} p(y_{n+1} = k | y_1, \dots, y_n) &= \int \cdots \int_{\Theta} p(y_{n+1} | (\lambda_1, \dots, \lambda_K)) p((\lambda_1, \dots, \lambda_K)) d\lambda_1 \dots d\lambda_K \\ &= \left(\frac{\Gamma(\sum_{j=1}^K \alpha_j + n_j)}{\prod_{j=1}^K \Gamma(\alpha_j + n_j)} \cdot \left(\frac{\Gamma(1 + \sum_{j=1}^K \alpha_j + n_j)}{\Gamma(a_k + n_k + 1) \prod_{j \neq k} \Gamma(\alpha_j + n_j)} \right)^{-1} \right) \\ &= \left(\frac{\Gamma(\sum_{j=1}^K \alpha_j + n_j)}{\Gamma(1 + \sum_{j=1}^K \alpha_j + n_j)} \cdot \left(\frac{\prod_{j=1}^K \Gamma(\alpha_j + n_j)}{\Gamma(a_k + n_k + 1) \prod_{j \neq k} \Gamma(\alpha_j + n_j)} \right)^{-1} \right) \\ &= \left(\frac{\Gamma(\sum_{j=1}^K \alpha_j + n_j)}{\Gamma(\sum_{j=1}^K \alpha_j + n_j) \cdot (\sum_{j=1}^K \alpha_j + n_j)} \cdot \left(\frac{\prod_{j=1}^K \Gamma(\alpha_j + n_j)}{\Gamma(a_k + n_k + 1) \prod_{j \neq k} \Gamma(\alpha_j + n_j)} \right)^{-1} \right) \\ &= \left(\frac{1}{(\sum_{j=1}^K \alpha_j + n_j)} \cdot \left(\frac{\prod_{j=1}^K \Gamma(\alpha_j + n_j)}{(a_k + n_k) \prod_{j=1}^K \Gamma(\alpha_j + n_j)} \right)^{-1} \right) \\ &= \frac{a_k + n_k}{\alpha + n} \end{aligned}$$

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## Data
# SCHOOL s=1
Y1 <- c(
  2.11, 9.75, 13.88, 11.3, 8.93, 15.66, 16.38, 4.54, 8.86, 11.94, 12.47, 11.11,
  11.65, 14.53, 9.61, 7.38, 3.34, 9.06, 9.45, 5.98, 7.44, 8.5, 1.55, 11.45, 9.73
)

# SCHOOL s=2
Y2 <- c(
  0.29, 1.13, 6.52, 11.72, 6.54, 5.63, 14.59, 11.74, 9.12, 9.43, 10.64, 12.28,
  9.5, 0.63, 15.35, 5.31, 8.49, 3.04, 3.77, 6.22, 2.14, 6.58, 1.11
)

# SCHOOL s=3
Y3 <- c(
  4.33, 7.77, 4.15, 5.64, 7.69, 5.04, 10.01, 13.43, 13.63, 9.9, 5.72, 5.16, 4.33
  , 12.9, 11.27, 6.05, 0.95, 6.02, 12.22, 12.85
)

## Initialization
Y <- list(Y1, Y2, Y3)
Ynames <- list("Y1", "Y2", "Y3")
```

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# Inputs:
# mu0 - first parameter mu of prior on mu prior
# tau0 - second parameter sigma^2 on mu prior
# a - first parameter a on sigma^{-2} prior
# b - second parameter b on sigma^{-2} prior
# Yi - samples
# Outputs
# gibbsIter - Function taking previous samples mu_t and sigma^2_t and returning
# next samples mu_{t+1} and sigma_{t+1}
iterFunc <- function(mu0, tau0, a, b, Yi) {
  gibbsIter <- function(mui, sigmai) {
    n <- length(Yi)
    muVar <- 1 / (n / sigmai + 1 / tau0)
    muMean <- muVar * (sum(Yi) / sigmai + mu0 / tau0)
    muNew <- rnorm(1, mean = muMean, sd = sqrt(muVar))
    aTilde <- a + n / 2
    bTilde <- b + 1 / 2 * sum((Yi - rep(muNew, n))^2)
    sigmaNew <- 1 / rgamma(1, shape = aTilde, rate = bTilde)
    c(muNew, sigmaNew)
  }
  gibbsIter
}
# Inputs:
# Yi - samples
# length - Number of samples to generate
# Outputs
# (mus, sigmas) - Vector with first length many items being the mu samples and
# remaining length many items the sigma samples
Gibbs <- function(Yi, length) {
  mu0 <- rnorm(1, mean = 5, sd = sqrt(4))
  sigma0 <- 1 / rgamma(1, shape = 1, rate = 4)
  iterYi <- iterFunc(mu0, sigma0, 1, 4, Yi)
  mus <- rep(1, length)
  sigmas <- rep(1, length)
  mus[1] <- mu0
  sigmas[1] <- sigma0
  iterYi(mu0, sigma0)
  for (i in 2:length) {
    t <- iterYi(mus[i - 1], sigmas[i - 1])
    mus[i] <- t[1]
    sigmas[i] <- t[2]
  }
  c(mus, sigmas)
}

# Perform samples
length <- 20000
a1 <- 1
b1 <- length
b2 <- 2 * length
a2 <- length + 1
mus <- list(1, 2, 3)
sigmas <- list(1, 2, 3)
for (i in 1:3) {
  samples <- Gibbs(Y[[i]], length)
  mus[[i]] <- samples[a1:b1]

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  sigmas[[i]] <- samples[a2:b2]
}

# Permutations we use for (d) and (e)
# I looked it up and there is a package for this but this is only six rows
permutations <- matrix(0, 6, 3)
permutations[1, 1:3] <- c(1, 2, 3)
permutations[2, 1:3] <- c(1, 3, 2)
permutations[3, 1:3] <- c(2, 1, 3)
permutations[4, 1:3] <- c(2, 3, 1)
permutations[5, 1:3] <- c(3, 1, 2)
permutations[6, 1:3] <- c(3, 2, 1)

## (a)
for (i in 1:3) {
  muMean <- mean(mus[[i]])
  sigmaMean <- mean(sigmas[[i]])
  print(paste("The posterior mean for mu of", Ynames[[i]], "is", muMean))
  print(paste("The posterior mean for sigma^2 of", Ynames[[i]], "is", sigmaMean))
}

## [1] "The posterior mean for mu of Y1 is 8.37153954399216"
## [1] "The posterior mean for sigma^2 of Y1 is 16.5007601742381"
## [1] "The posterior mean for mu of Y2 is 6.41838689526267"
## [1] "The posterior mean for sigma^2 of Y2 is 20.5232504323136"
## [1] "The posterior mean for mu of Y3 is 7.15752397947134"
## [1] "The posterior mean for sigma^2 of Y3 is 15.1952568219661"

## (b)
for (i in 1:3) {
  muQuant <- quantile(mus[[i]], probs = c(0.025, 0.975))
  sigmaQuant <- quantile(sigmas[[i]], probs = c(0.025, 0.975))
  print(paste(
    "The empirical 95% confidence interval for mu of", Ynames[[i]],
    "is(", muQuant[1], ",", muQuant[2], ")")
  ))
  print(paste(
    "The empirical 95% confidence interval for sigma^2 of",
    Ynames[[i]], "is (", sigmaQuant[1], ",", sigmaQuant[2], ")")
  ))
}

## [1] "The empirical 95% confidence interval for mu of Y1 is( 6.83355671274148 , 9.73211055217593 )"
## [1] "The empirical 95% confidence interval for sigma^2 of Y1 is ( 9.2373888642346 , 29.2152700323206 )"
## [1] "The empirical 95% confidence interval for mu of Y2 is( 4.85173789870027 , 7.91677692218941 )"
## [1] "The empirical 95% confidence interval for sigma^2 of Y2 is ( 11.4912407849963 , 36.5299422767055 )"
## [1] "The empirical 95% confidence interval for mu of Y3 is( 5.58592916404238 , 8.6044994144952 )"
## [1] "The empirical 95% confidence interval for sigma^2 of Y3 is ( 8.02663143318418 , 28.5083091969154 )"

## (c)
for (i in 1:6) {
  perm <- permutations[i, 1:3]
  t <- 0
  # Check and tally for which samples this holds true
  for (j in 1:length) {

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    if (mus[[perm[1]]][j] < mus[[perm[2]]][j]) {
      if (mus[[perm[2]]][j] < mus[[perm[3]]][j]) {
        t <- t + 1
      }
    }
  }
}
# Normalize and print
t <- t / length
cat(sprintf("P(mu%d < mu%d < mu%d) = %f\n", perm[1], perm[2], perm[3], t))
}

## P(mu1 < mu2 < mu3) = 0.012900
## P(mu1 < mu3 < mu2) = 0.007550
## P(mu2 < mu1 < mu3) = 0.106300
## P(mu2 < mu3 < mu1) = 0.633150
## P(mu3 < mu1 < mu2) = 0.016000
## P(mu3 < mu2 < mu1) = 0.224100

## (d)
# This is an application of the indirect Monte Carlo approximation from lecture
# 5.
for (i in 1:6) {
  perm <- permutations[i, 1:3]
  t <- 0
  for (j in 1:length) {
    # Very similar to (c) though we sample each distribution once
    samples <- 1:3
    for (k in 1:3) {
      mu <- mus[[perm[k]]][j]
      sigma <- sigmas[[perm[k]]][j]
      samples[k] <- rnorm(1, mu, sqrt(sigma))
    }
    if (samples[1] < samples[2]) {
      if (samples[2] < samples[3]) {
        t <- t + 1
      }
    }
  }
}
t <- t / length
cat(sprintf("P(~Y%d < ~Y%d < ~Y%d) = %f\n", perm[1], perm[2], perm[3], t))
}

## P(~Y1 < ~Y2 < ~Y3) = 0.113750
## P(~Y1 < ~Y3 < ~Y2) = 0.115850
## P(~Y2 < ~Y1 < ~Y3) = 0.189150
## P(~Y2 < ~Y3 < ~Y1) = 0.248500
## P(~Y3 < ~Y1 < ~Y2) = 0.141150
## P(~Y3 < ~Y2 < ~Y1) = 0.192900

## (e)
t <- 0
for (i in 1:length) {
  if (mus[[1]][i] > mus[[2]][i]) {
    if (mus[[1]][i] > mus[[3]][i]) {
      t = t + 1
    }
  }
}
}

```

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}
t <- t / length
cat(sprintf("P(mu1 < mu2, mu1 < mu3) = %f\n", t))

## P(mu1 < mu2, mu1 < mu3) = 0.857250

## (f)
t <- 0
for (j in 1:length) {
  samples <- 1:3
  for (k in 1:3) {
    mu <- mus[[k]][j]
    sigma <- sigmas[[k]][j]
    samples[k] <- rnorm(1, mu, sqrt(sigma))
  }
  if (samples[1] > samples[2]) {
    if (samples[2] > samples[3]) {
      t <- t + 1
    }
  }
}
t <- t / length
cat(sprintf("P(~Y1<~Y2, ~Y1<~Y3) = %f\n", t))

## P(~Y1<~Y2, ~Y1<~Y3) = 0.198450

```